

Kohler AI Bathroom Designer and Planner

AI Prompts, System-Level Prompt Contracts & Workflows

This PDF is committed to the repository detailing all AI prompts, system instructions, and workflows used to build your solution.

Rather than relying on a single AI, I run a multi-agent feedback loop between ChatGPT and Claude to stress-test alternate approaches, using human-in-the-loop orchestration to filter for the best technical strategy.

List of AI prompts used to design the system, choosing right tech stack and getting familiar with the track:

- *Can you tell me some similar products I can try to get more familiar about the track*
- *What are the different solution approaches for track 1 given I want AI intelligence that can actually help improve product recommendation and generate a 2D representation of the bathroom layout with selected products.*
- *The Knowledge Graph (The Rules Engine): i think neo4j is not a wrong choice i should use it for simplicity and industry dominance for KG*
- *Does the current solution address all above mentioned situations??*
- *Explain the whole pipeline once again in easy language also state it's features what it provides to end users, any flaws in the system, also an architectural design of the system*
- *Is there a fall back if there is no catalog under derired budget and theme*
- *Fix all the 4 flaws in the pipeline*
- *One thing that is STILL not fully solved*
There is one important weakness remaining:

Accessories are added after optimization.

Currently: (demonstrating an example for the case in the prompt) So your optimizer thought it was under budget, but the real bundle isn't.

Your own comment recognizes this.

For a stronger final version, make accessories part of the optimization itself: (giving proposed solution in prompt), Think wisely before applying the changes... will it be a good design and wont create nay problems??

- *so suppose i chose japanse zen bathroom design, and if nothing matches my budget whats the reponse i get*
- *fix this, with proper reasoning*
- *but does this it intelligently places the toilet and vanity and shower with respect to the door so it looks beautiful and aesthetic, also does the 2d layout includes additional things like electrical proximity?? is it really working like an interior aesthetic planner or just a math algo performing chechks if any thing is out of the carpet area or overlapping*
- *Just answer one question that: does it take the door opening radius as input, talking about the last final pipeline??*

List of AI prompts and instructions used to build the repository and code the solution:

System Instruction:

Implement the Kohler AI Bathroom Designer according to the specification.

Work incrementally through the phases defined in the spec:

- 1. Foundation and project structure*
- 2. Synthetic catalog and evaluation data*
- 3. Neo4j knowledge graph*
- 4. Hybrid product recommendation*
- 5. PuLP bundle optimization*
- 6. Multimodal spatial extraction*
- 7. Deterministic geometry validation*
- 8. Layout candidate generation and refinement*
- 9. Deterministic SVG rendering*

10. Flask frontend and API integration
11. End-to-end integration and evaluation
12. Testing and demo polish

Do not start by blindly generating the entire project at once.

First inspect the repository and existing code, then create a concise implementation plan.

Preserve useful existing functionality and architecture where possible.

After each major phase:

- run the relevant tests
- inspect failures
- fix regressions
- continue to the next phase

Do not consider the task complete just because the files have been created. The complete end-to-end demo must actually run locally.

The final flow must work:

Bathroom image + user requirements

- spatial extraction
- product KG + semantic retrieval
- hybrid recommendation
- PuLP bundle optimization
- Top 3 bundles
- 3–5 layout candidates
- deterministic geometry validation
- refinement of invalid layouts
- best valid layout
- deterministic SVG
- Flask interactive frontend

Also implement the product replacement follow-up flow described in IMPLEMENTATION_SPEC.md.

AI prompts used:

- *Read IMPLEMENTATION_SPEC.md completely before making any changes. Then inspect the entire repository and understand the existing code. Read the instruction provided carefully.*
- *create a folder from scratch and place the code files*
- *i have 21.13 dollar HuggingFace credits can i use it some how intead of buying openai api but dont you think my output quality will degrade*
- *Glve me a separate Readme file for github, should include the following in order: Project title: Kohler AI Bathroom Planner and Designer, tech stack in the format:{ for example: [\[!\[OpenEnv\]\(https://img.shields.io/badge/OpenEnv-compliant-4CAF50?style=flat-square\)\]\(https://github.com/meta-pytorch/OpenEnv\)](https://img.shields.io/badge/OpenEnv-compliant-4CAF50?style=flat-square) }, problem statement, the solution, the imapct in business and competion in the field of ai in manufacturing + recommendation engine and interior planning, salient feature and constraints considered, system architecture, system archtucture for each module in order, repo structure (with short comments defining function of each file), setup and usage, author: Satyam Kumar , DONT RUSH while drawing syatem architure for the project or for each module, double check its correctness, generate the file incrementally*
- *verify once agin with previous chats and with the current repo structure if everything is fine*

NOTE: Debugging-related AI interactions containing large terminal logs, stack traces, and diagnostic outputs are omitted to keep this document focused on the final production prompts and AI workflows.